

Topic 9.1
Predictive Models interpretability

Introduction to Predictive Models Interpretability

By Dr Lee How Chinh

Objectives

- Define the predictive model interpretability.
- Explain the importance of predictive model interpretability.

Introduction to Predictive Models Interpretability

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- Interpretability refers to the degree to which a human can understand the cause of a decision made by a predictive or machine-learning model.
- Modern machine learning algorithms, such as neural networks and ensemble methods like random forests and gradient boosting, can learn complex relationships in data.
- However, the resulting predictive models can be highly complex, with millions of parameters or intricate decision boundaries.
- This complexity often makes it challenging for humans to understand how the model arrives at its predictions, leading to black-box models.
- Black-box models provide accurate predictions but lack transparency in explaining why specific decisions are made, hindering trust and understanding.

Importance of interpretability

Importance of interpretability for various stakeholders

- The interpretability of machine learning models is crucial for a range of stakeholders involved in decision-making processes, including regulators, executives, data scientists, and domain experts.
- Regulators need model interpretability to ensure that decisions made by the model comply with laws and regulations, especially regarding fairness and non-discrimination.
- Executives require an understanding of black-box models to justify decisions made by these models logically and ethically.
- Data scientists and domain experts need interpretability to detect biases, extract insights from models, and debug errors or unexpected predictions effectively.

Importance of interpretability for various stakeholders

- Importance of Interpretability:
  - Facilitates understanding, debugging, and auditing of model predictions.
  - Aids in bias detection to ensure fair decision-making.
  - Enables robustness checks to maintain output stability despite small input changes.
  - Provides recourse for individuals affected by model predictions.
- Trade-off: Balance between model performance and interpretability. Higher performance often sacrifices interpretability and vice versa.

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Topic 9.2
Predictive Models interpretability

Methods for Predictive Model Interpretability

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Objectives

- State the methods for predictive model interpretability.
- Choose and apply the suitable model interpretability tools.

Methods for Model Interpretability

Methods for Model Interpretability

- Model interpretability encompasses two main categories: inherently interpretable models and post-hoc explanations.
- **Inherently interpretable models** are designed with transparency in mind from the outset, directly incorporating interpretability into their structure. Examples include generalised linear models, which offer straightforward explanations through their coefficient estimates.
- **Post-hoc explanations**, on the other hand, are techniques applied after a model is trained to explain its predictions. These methods aim to provide insights into the inner workings of complex models, making them more understandable to humans.

Global and Local Interpretability

- Within post-hoc interpretability, techniques can be broadly categorised into **global and local interpretability** methods.
- **Global interpretability techniques** strive to reveal the model's overall behaviour across the entire dataset. They reveal overarching patterns, such as which features most influence driving predictions or how each feature impacts the model's overall output.
- **Local interpretability methods**, conversely, zoom in on individual predictions, offering explanations tailored to specific instances. By focusing on individual cases, these techniques provide insights into why a particular prediction was made, offering valuable context and understanding.

Global Interpretability Techniques

Global Interpretability: Variable Importance

- Variable importance tables and partial dependence (PD) plots are two commonly used global interpretability techniques, and you can access them in Model Studio.
- Variable Importance Tables: These tables provide insight into the importance of each feature in the predictive model. By indicating the statistical contribution of each feature, stakeholders can understand which variables have the most influence on the model's predictions. Various methods exist for calculating feature importance, such as fitting a global surrogate decision tree model or using permutation-based feature importance.

Global Interpretability: PD Plots

- Partial Dependence Plots (PD): PD plots visually represent the relationship between a specific feature and the model's average prediction. By varying the values of one feature while keeping others constant and observing the resulting change in predictions, stakeholders can understand how a feature influences the model's output on average across the entire dataset.
- Refer to How to Explain Your Black-Box Models in SAS Viya for the steps to produce PD plots.

Local Interpretability Techniques

Local Interpretability

- Local interpretability methods aim to explain individual predictions of machine learning models.
- The common techniques are Individual conditional expectation (ICE) plots, LIME (Local Interpretable Model-Agnostic Explanations) and Shapley Values.
- These techniques focus on explaining why a specific prediction was made for a particular instance.

Local Interpretability: ICE plots

- **Individual Conditional Expectation (ICE) Plots**: Unlike PD plots that focus on average effects, ICE plots provide insights into how a feature affects predictions for individual observations.
- By plotting the predictions for each individual observation as the feature of interest varies, stakeholders can see how the impact of a feature differs across different data points.
- ICE plots allow for a more granular understanding of feature effects, revealing variations in predictions among different individuals.
- The procedure for producing ICE plots is the same as that for creating PD plots, except that ICE plots are performed only on a single observation.

Local Interpretability: LIME

- LIME (Local Interpretable Model-Agnostic Explanations) is a technique used to explain individual predictions of any machine-learning model by building a white-box local surrogate model.
- It works by sampling the feature space around a specific observation and then training a sparse linear regression model, such as LASSO, on this generated sample.
- The surrogate model approximates the behaviour of the black-box model locally, making it easier to understand.

Local Interpretability: Shapley Values

- Shapley values are a method for explaining individual predictions by averaging across all permutations of the features.
- Unlike LIME coefficients, Shapley values control for variable interactions, making them more robust in capturing the contribution of each feature to the prediction.
- This allows for a more comprehensive understanding of how each feature contributes to the model's predictions.

Local Interpretability: Kernel SHAP

- Kernel SHAP is a method for computing Shapley values based on synthetic data generation and weighted regression.
- It provides an efficient way to compute Shapley values by generating synthetic data that follows the distributions of the input data set and then scoring this data using the machine learning model.
- Kernel SHAP is implemented in SAS Viya, offering users a computationally efficient approach to compute Shapley values.
- By leveraging synthetic data generation and weighted regression, Kernel SHAP provides accurate explanations for individual predictions, making it a valuable tool for model interpretability in SAS Viya.

Lesson Summary

- Many types of predictive models that can be highly flexible and learn complex relationships in data are black box models.
- However, the interpretability of the predictive models is crucial after the prediction process, such as auditing, bias detection, and assisting the affected individuals.
- There are two categories for predictive model interpretability: inherently interpretable models and post-hoc explanations.
- This topic focuses on post-hoc explanation methods. For global interpretability techniques, there are variable importance and PD plots. Local interpretability methods: ICE plots, LIME and Kernel SHAP.

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How does a Model Studio produce the global and local interpretability plots?

What do the plots look like?



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